

Source Code Management

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Slot: L25+L26

What is Git?

Git is a distributed version control system used to track changes in source code. It helps developers manage versions, collaborate with others, and revert to previous states if needed.

What is Version Control?

Version control is a system that records changes to files over time. It allows developers to track history, collaborate safely, and avoid losing work.

What is GitHub?

GitHub is a web-based platform that hosts Git repositories. It enables collaboration, sharing code, and managing projects online.

Difference Between Git and GitHub

Git is a tool; GitHub is a platform

- Git works locally; GitHub works online
- Git tracks changes; GitHub hosts repositories
- Git doesn't require internet; GitHub does
- Git is software; GitHub is a service

Basic Git Commands

git init: Initializes a new Git repository

```
anuj-preetham@annihilus:~/Desktop/scm$ git init
Initialized existing Git repository in /home/anuj-preetham/Desktop/scm/.git/
anuj-preetham@annihilus:~/Desktop/scm$
```

git status: Shows current status of files

```
anuj-preetham@annihilus:~/Desktop/scm$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    10mar/
    24mar/
    CN assignment 1.pdf
    Git Commands II (1).pdf
    Git Commands II.pdf
    file
    filename
    nano-cheatsheet.pdf
    scm lab file.odt
    scm lab file.pdf

nothing added to commit but untracked files present (use "git add" to track)
anuj-preetham@annihilus:~/Desktop/scm$
```

git add .: Adds all files to staging area

(no output message printed here, command executes and returns nothing to terminal prompt)

git commit -m "message": Saves changes with a message

```
anuj-preetham@annihilus:~/Desktop/scm$ git commit -m "SCM Lab Assignment March 31"
[main 0e7916d] SCM Lab Assignment March 31
18 files changed, 24 insertions(+)
create mode 100644 10mar/add.py
create mode 100644 10mar/display.html
create mode 100644 10mar/div.java
create mode 100644 10mar/multi.cpp
create mode 100644 10mar/sub.c
create mode 100644 24mar/file1.txt
create mode 100644 24mar/file2.txt
create mode 100644 24mar/file3.txt
create mode 100644 24mar/test.txt
create mode 100644 24mar/vi editor tasks.pdf
create mode 100644 CN assignment 1.pdf
create mode 100644 Git Commands II (1).pdf
create mode 100644 Git Commands II.pdf
create mode 100644 file
create mode 100644 filename
create mode 100644 nano-cheatsheet.pdf
create mode 100644 scm lab file.odt
create mode 100644 scm lab file.pdf
anuj-preetham@annihilus:~/Desktop/scm$
```

git add .: Adds all files to stag

(no output message printed here, com

git commit -m "message": S

git push: Uploads commits to

git pull: Fetches latest chang

git push: Uploads commits to GitHub

```
anuj-preetham@annihilus:~/Desktop/scm$ git push
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 16 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (17/17), 2.17 MiB | 239.60 KiB/s, done.
Total 17 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:legendracer/scm.git
   3e0ab4b..0e7916d  main -> main
anuj-preetham@annihilus:~/Desktop/scm$
```

git pull: Fetches latest changes

```
anuj-preetham@annihilus:~/Desktop/scm$ git pull
Already up to date.
anuj-preetham@annihilus:~/Desktop/scm$
```

git log: Shows commit history

```
anuj-preetham@annihilus:~/Desktop/scm$ git log
commit 0e7916d48ba5565ec62a35103e115f0b4d6b9131 (HEAD -> main, origin/main)
Author: Anuj Preetham <anujpreetham@gmail.com>
Date: Tue Mar 31 14:22:01 2026 +0530

    SCM Lab Assignment March 31

commit 3e0ab4bee92f335f1739be9e52ad87014fda95b6
Author: Anuj Preetham <anujpreetham@gmail.com>
Date: Tue Feb 24 13:48:28 2026 +0530

    Initial commit
anuj-preetham@annihilus:~/Desktop/scm$
```

git branch: Lists branches

```
anuj-preetham@annihilus:~/Desktop/scm$ git branch
* main
anuj-preetham@annihilus:~/Desktop/scm$
```

git checkout branch-name: Switches branch

```
anuj-preetham@annihilus:~/Desktop/scm$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
anuj-preetham@annihilus:~/Desktop/scm$
```